ALL MEDIA ARE ORIGINAL!

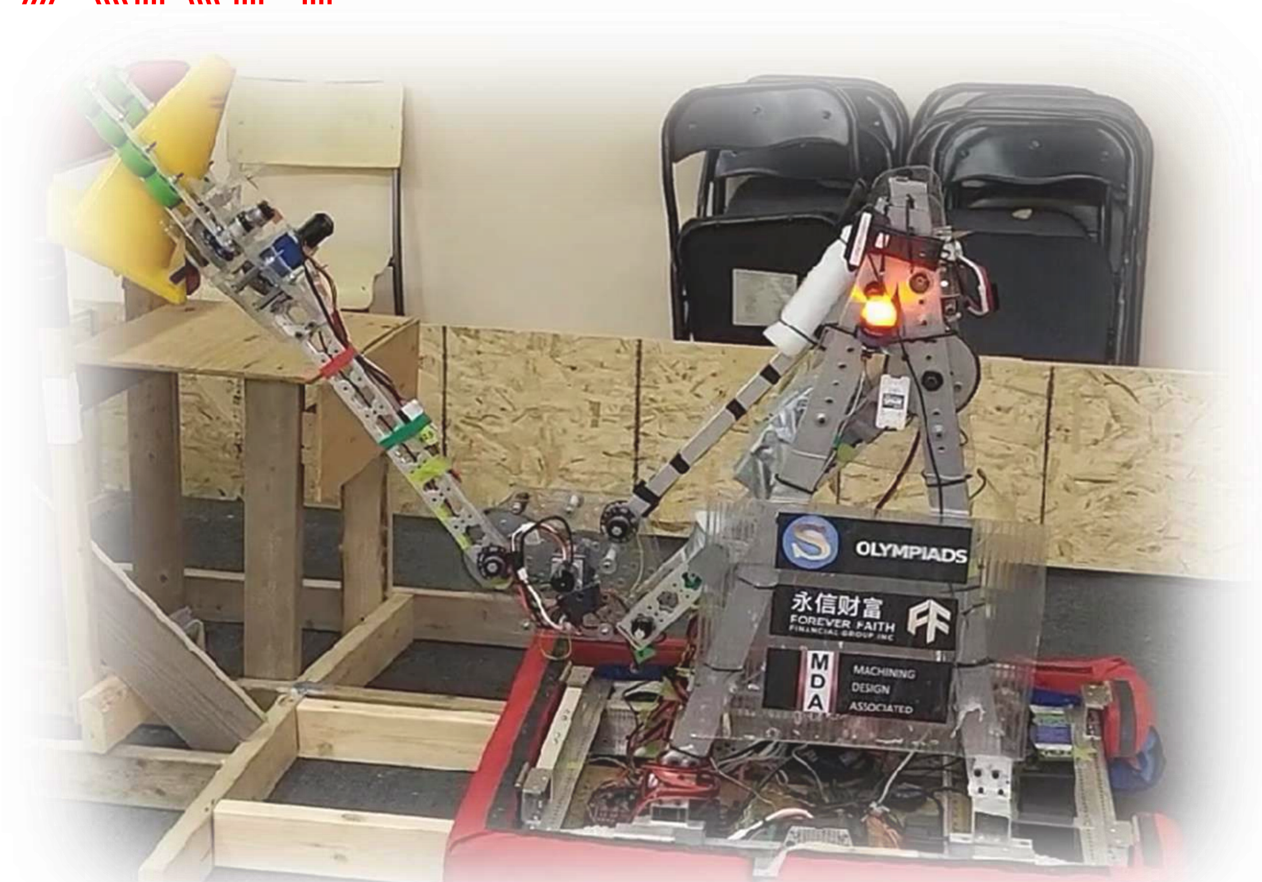
ROBIN YAN

PORTFOLIO

WORK IN PROGRESS

INVERSE KINEMATICS ARM

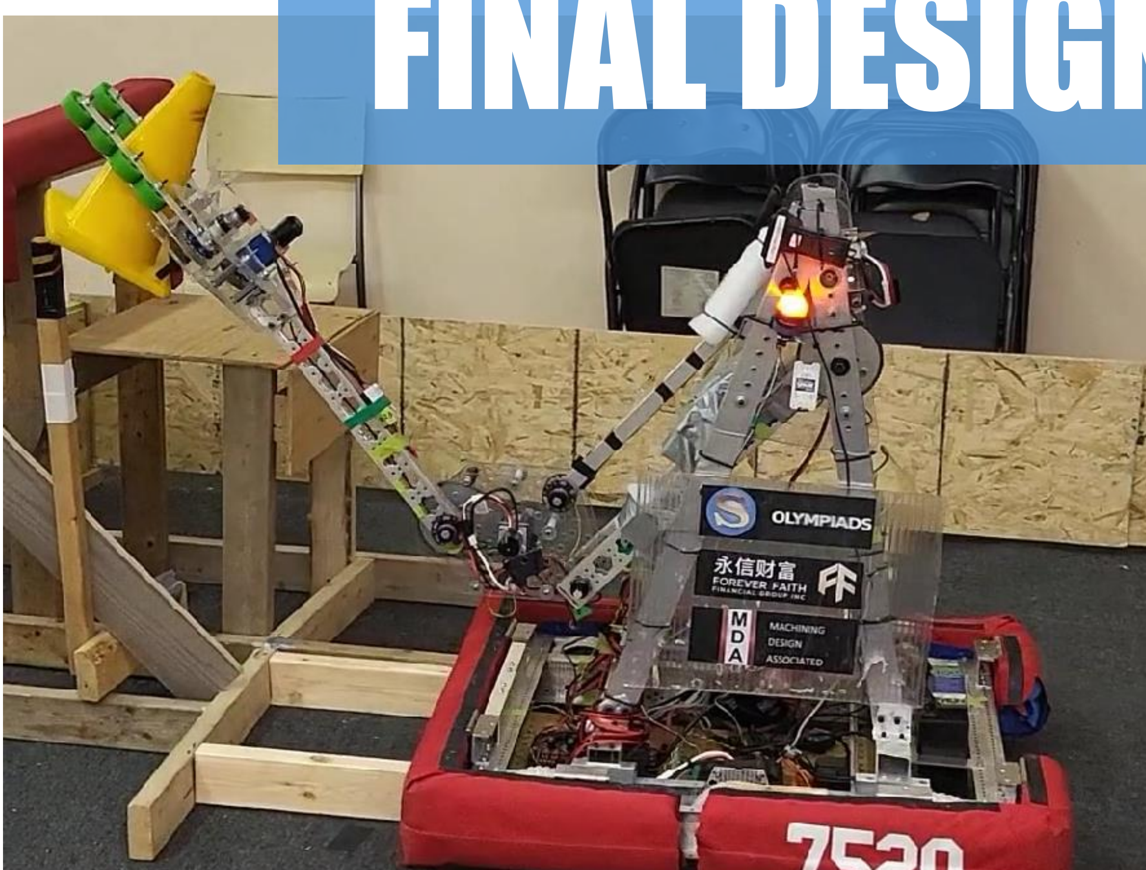
WORK IN PROGRESS!

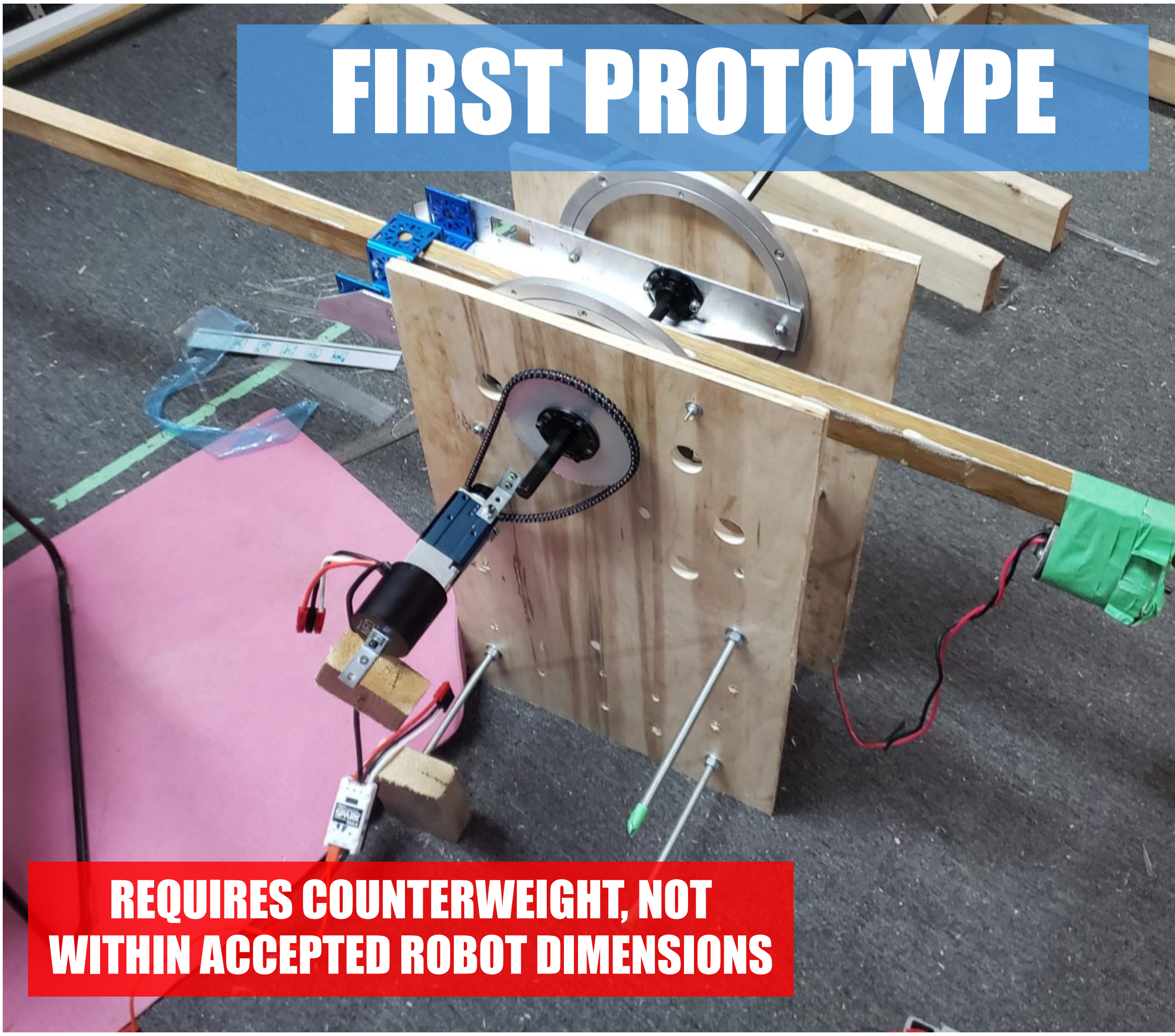


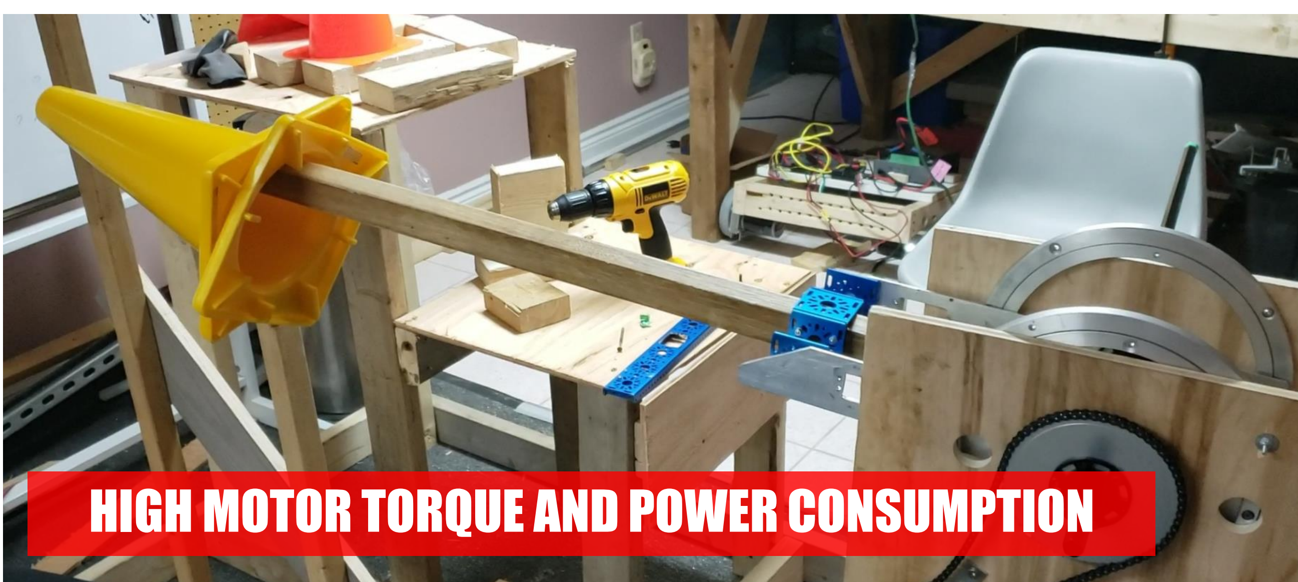
FRC 2023 ROBOT

ENGINEERING PROCESS

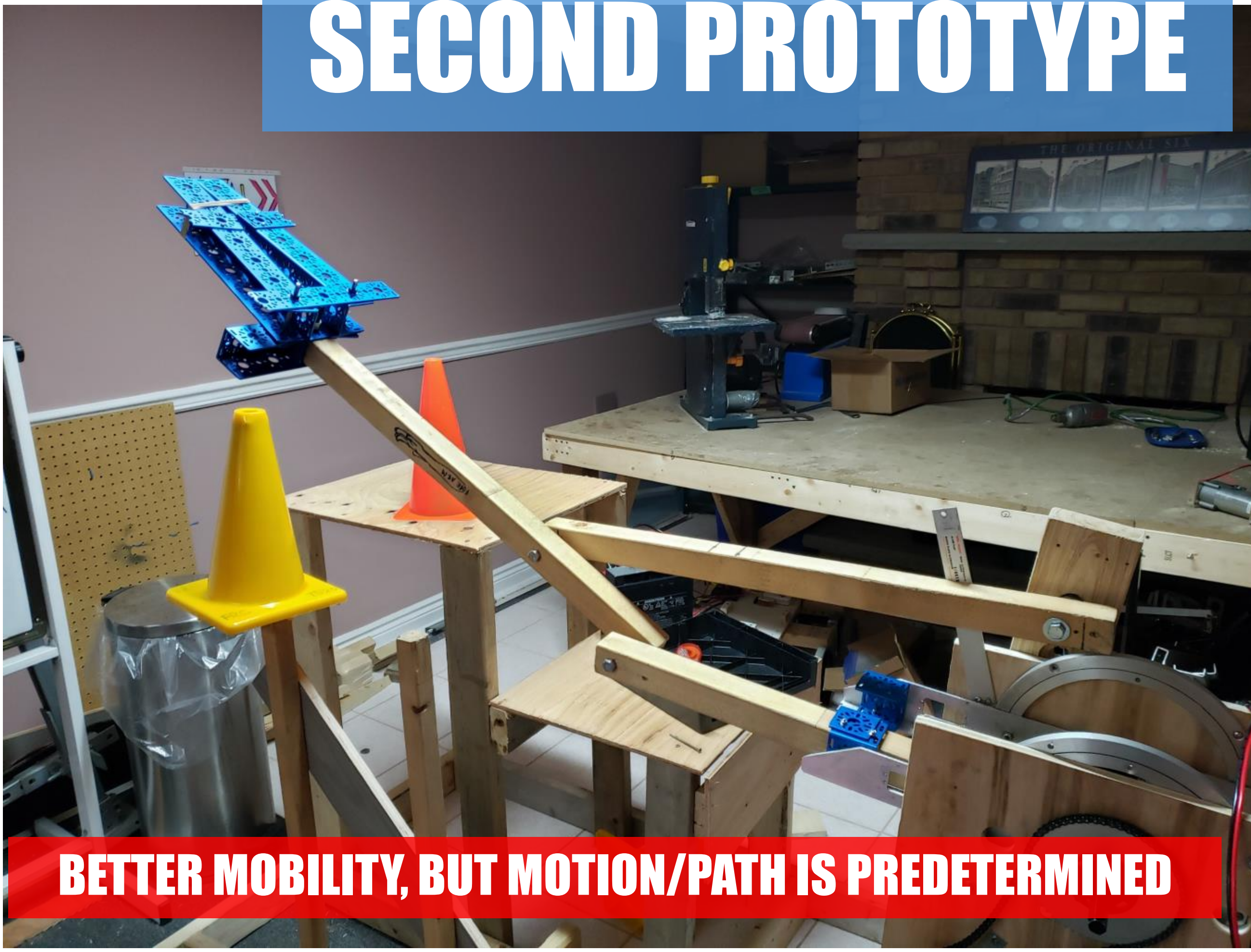
PREVIEW OF FINAL DESIGN

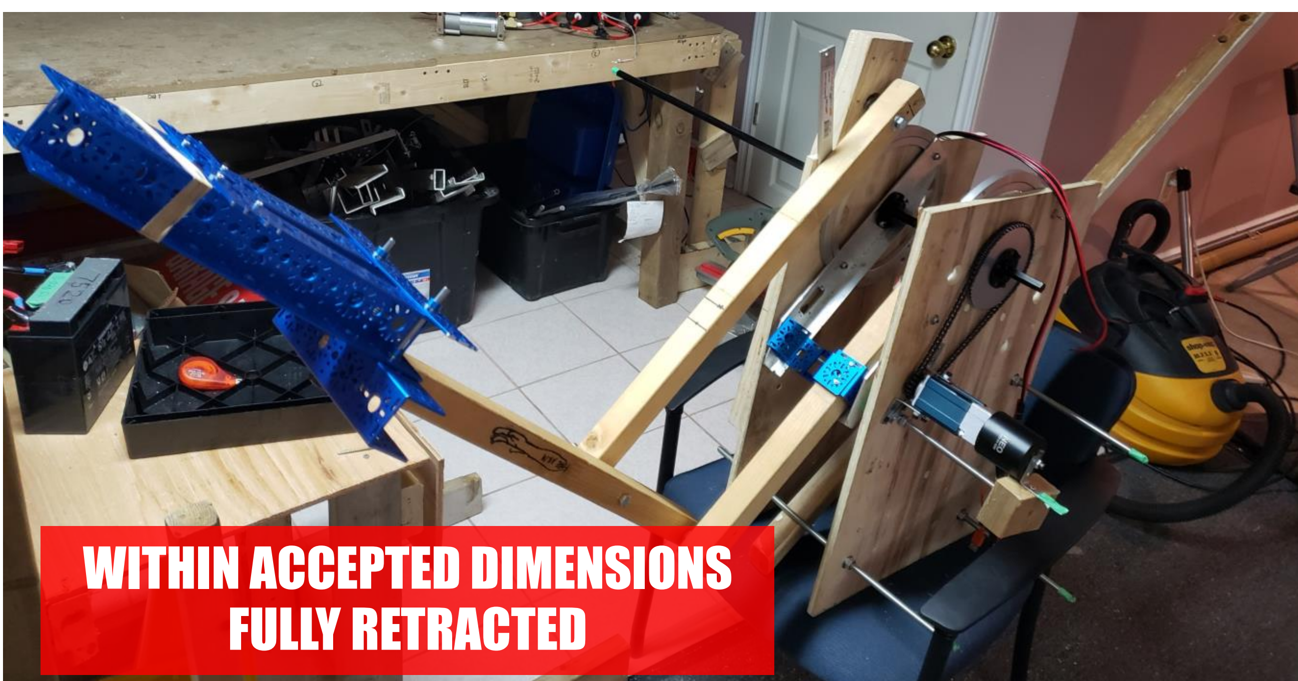


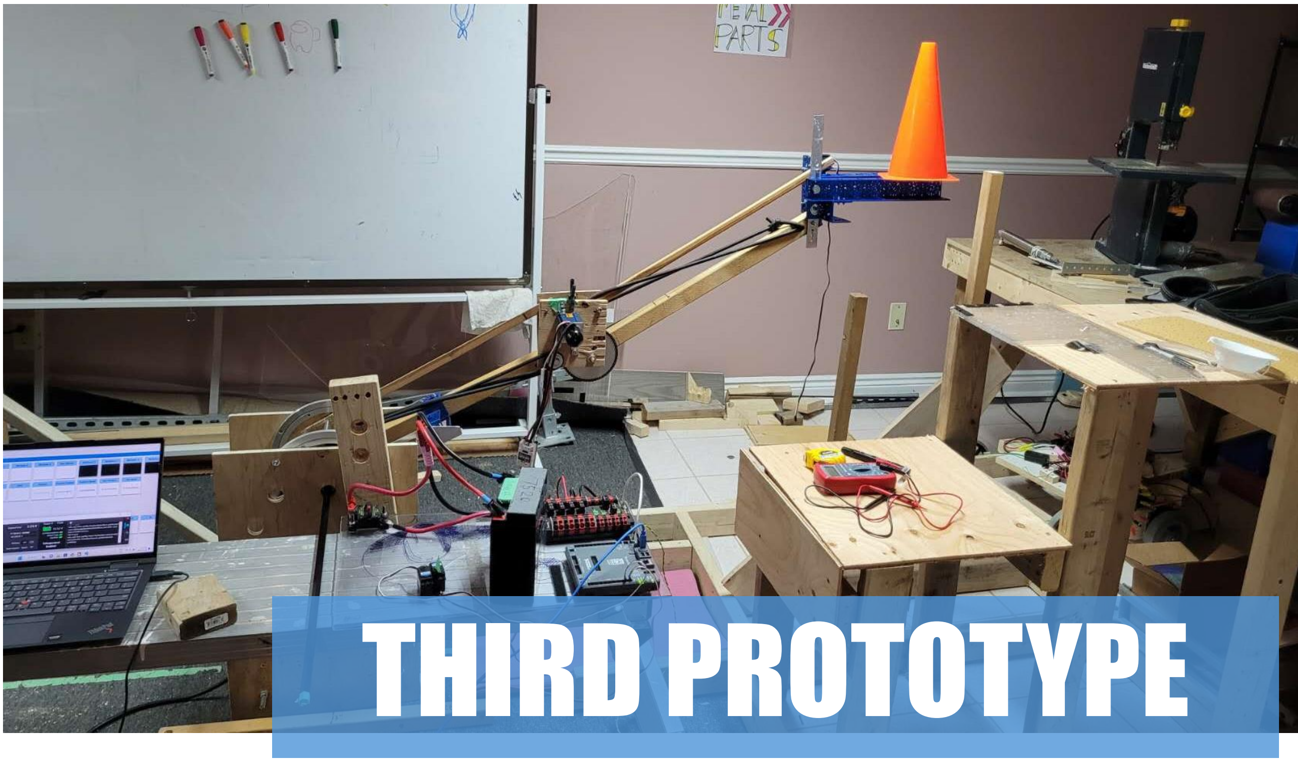

FIRST PROTOTYPE
REQUIRES COUNTERWEIGHT, NOT WITHIN ACCEPTED ROBOT DIMENSIONS


HIGH MOTOR TORQUE AND POWER CONSUMPTION

SECOND PROTOTYPE

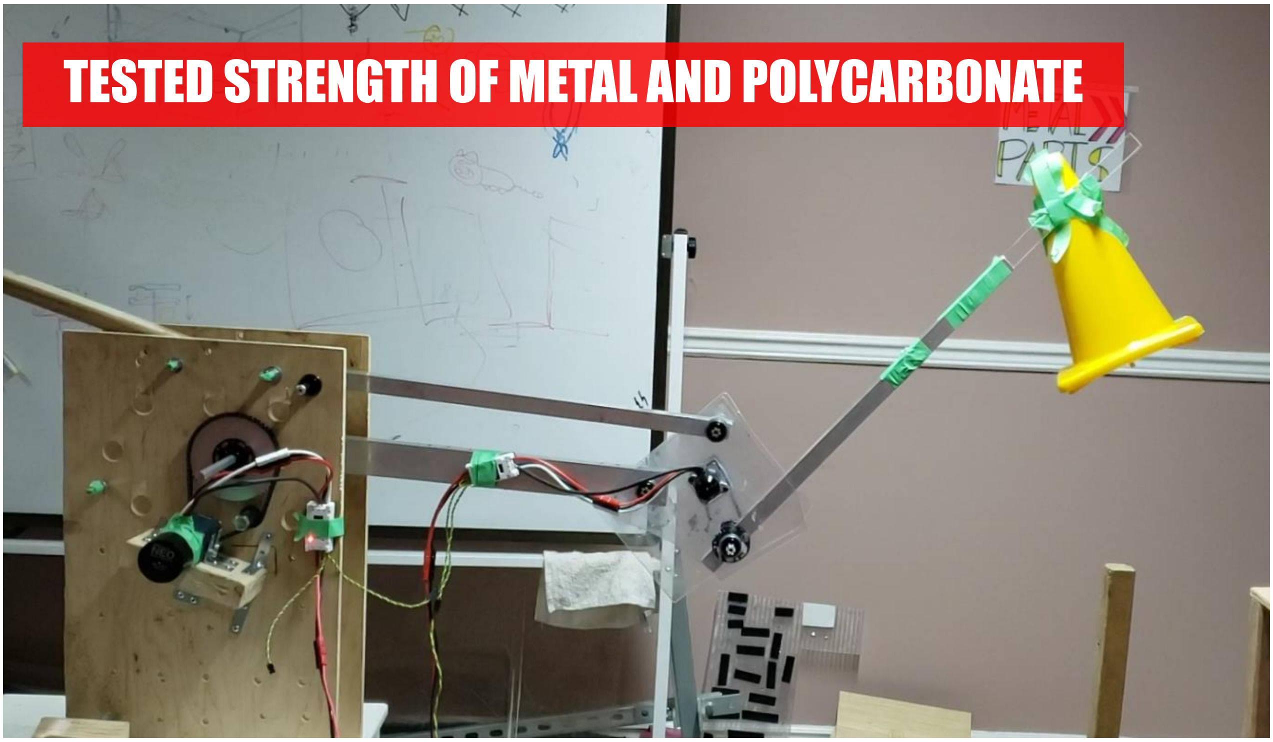





THIRD PROTOTYPE


ADDED PARALLEL-BARS AND 2nd DEGREE OF FREEDOM
NEW MECHANICAL RESTRICTIONS & NOT WITHIN ACCEPTED DIMENSIONS

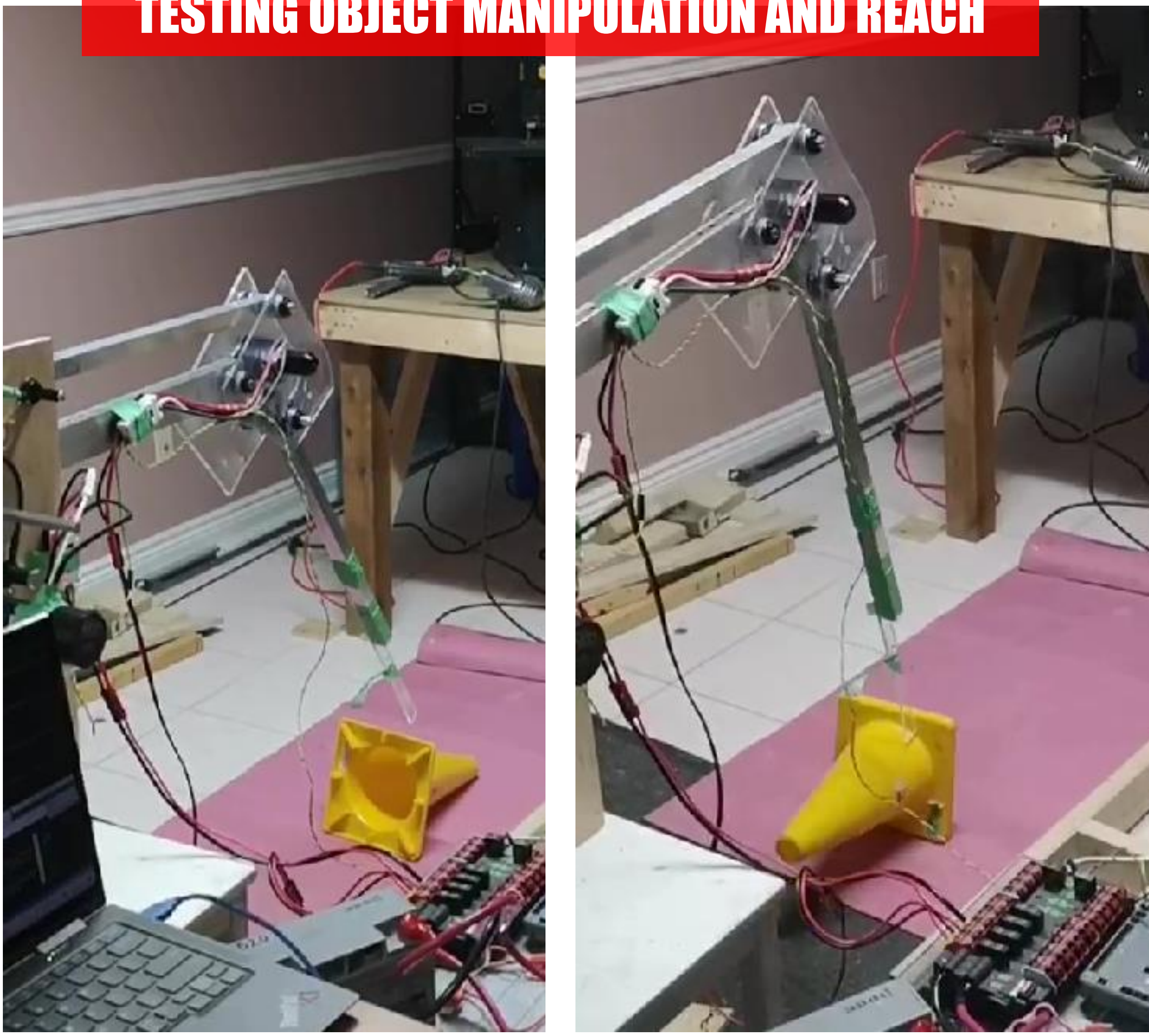
FOURTH PROTOTYPE

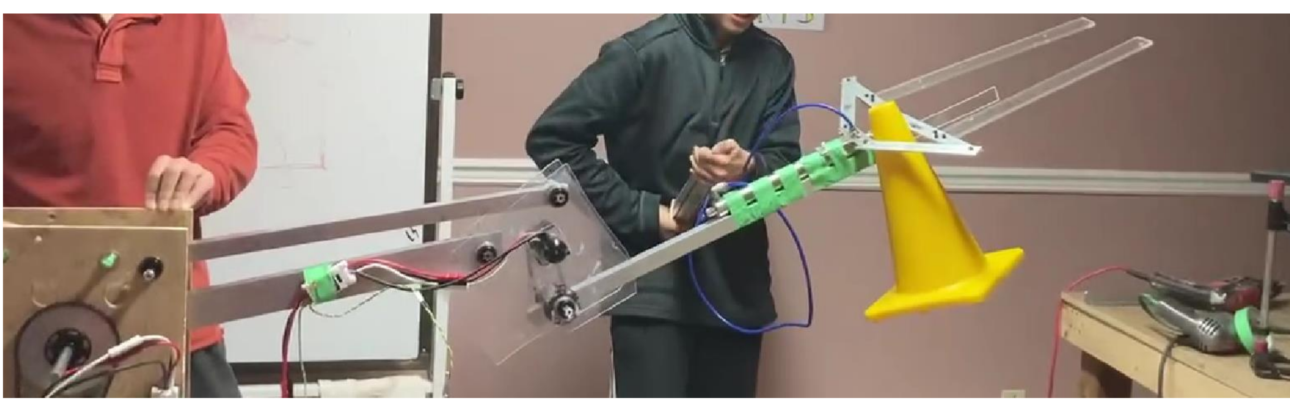


TESTED STRENGTH OF METAL AND POLYCARBONATE

BETTER ANGLES AND ACCESSIBILITY TO MORE PONTS OF INTEREST

TESTING OBJECT MANIPULATION AND REACH

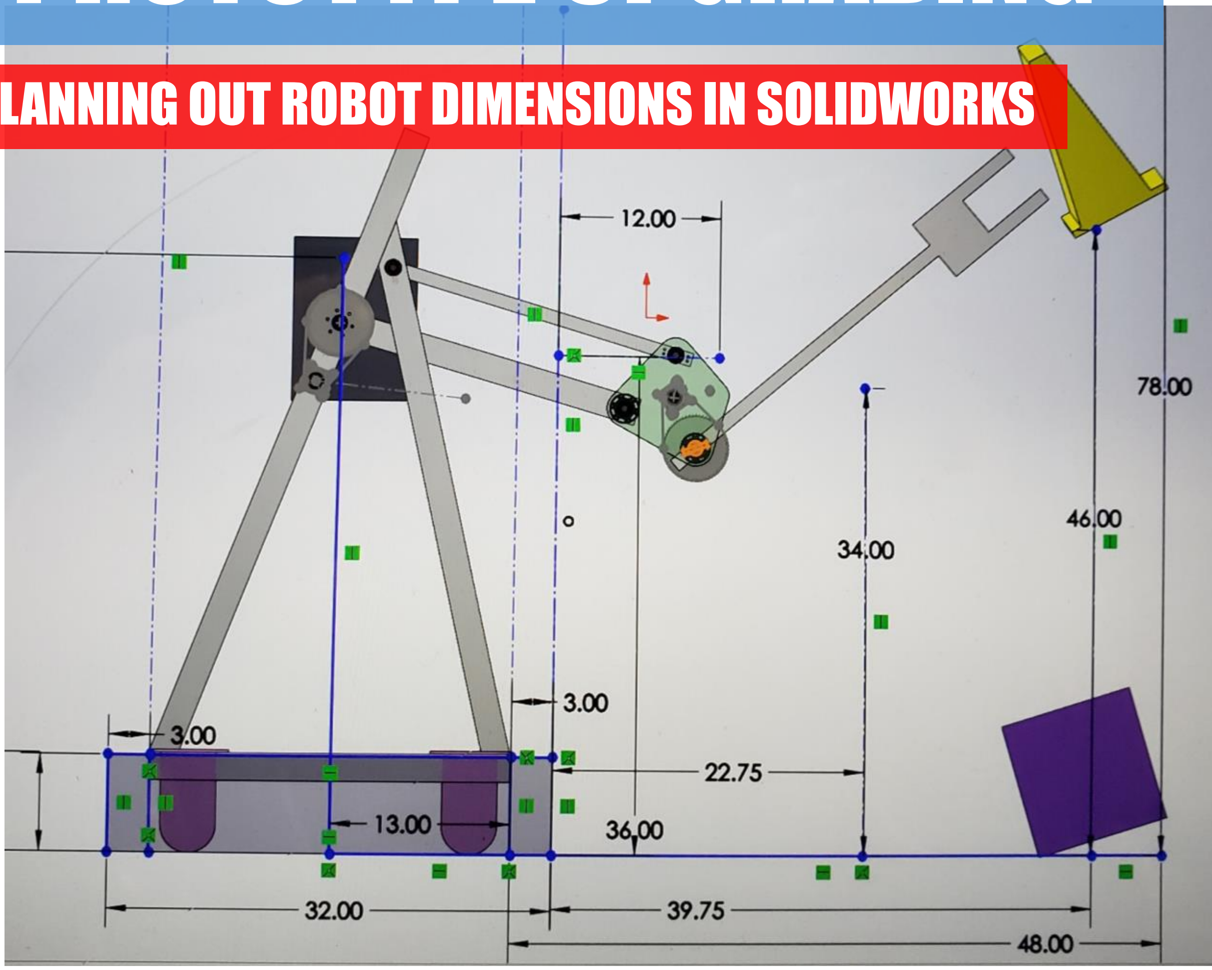


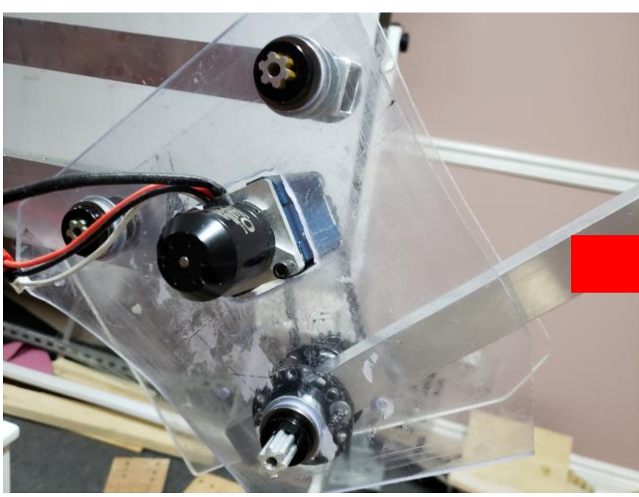


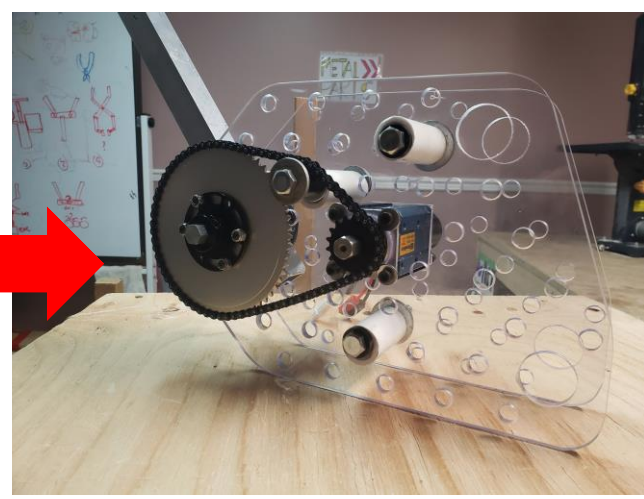
VALIDATING ARM FUNCTION AND CURRENT REQUIRED TO COUNTER GRAVITY

PROTOTYPE UPGRADING

PLANNING OUT ROBOT DIMENSIONS IN SOLIDWORKS

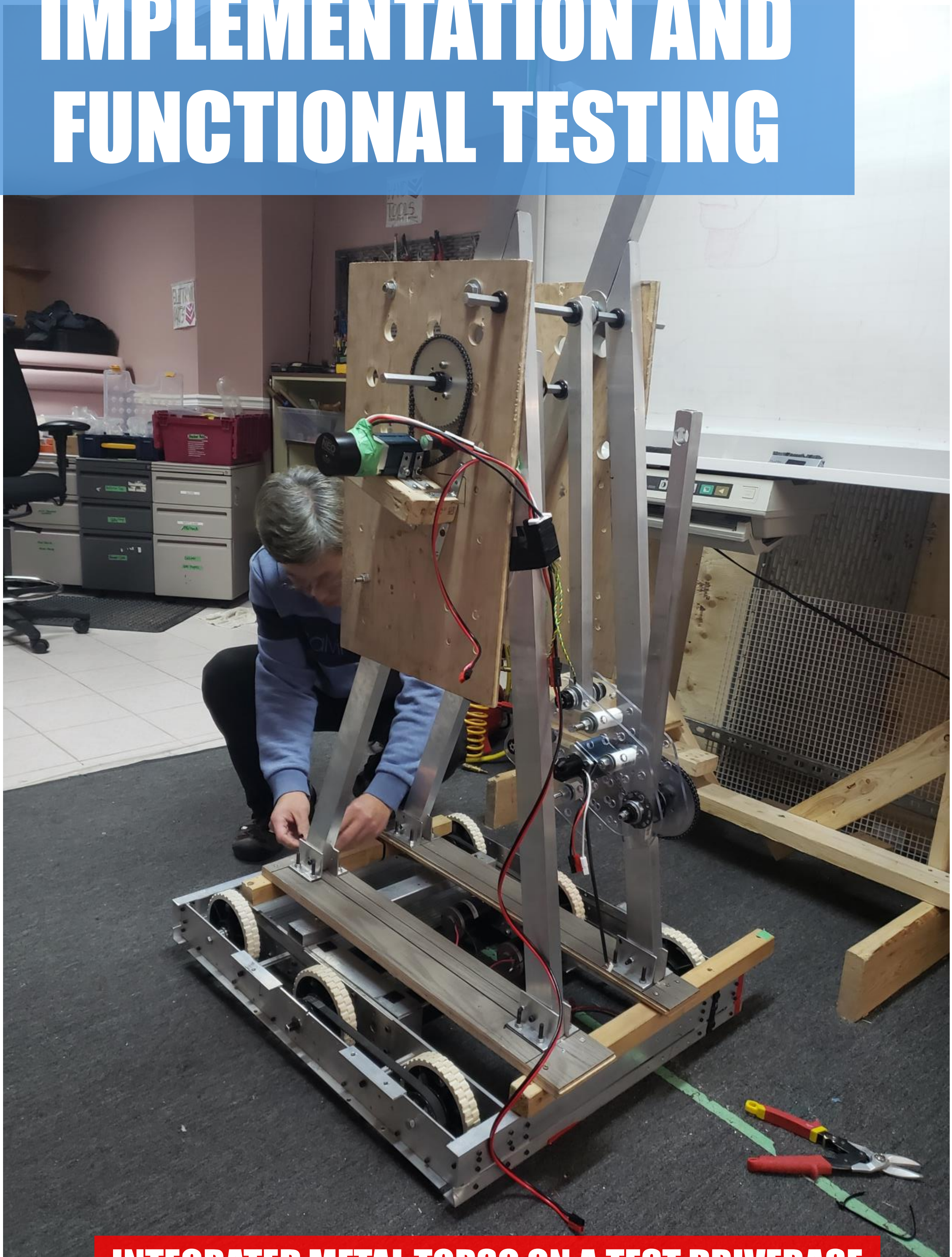




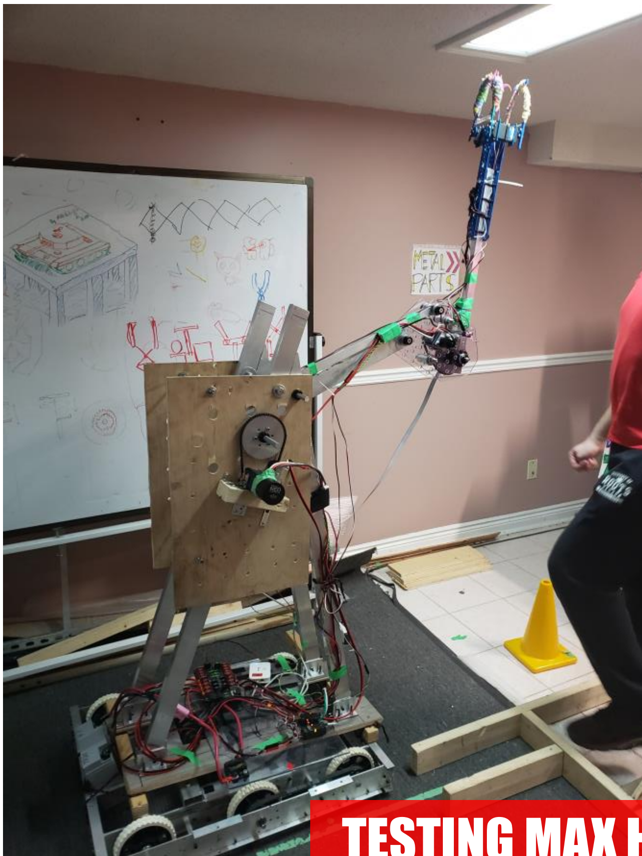


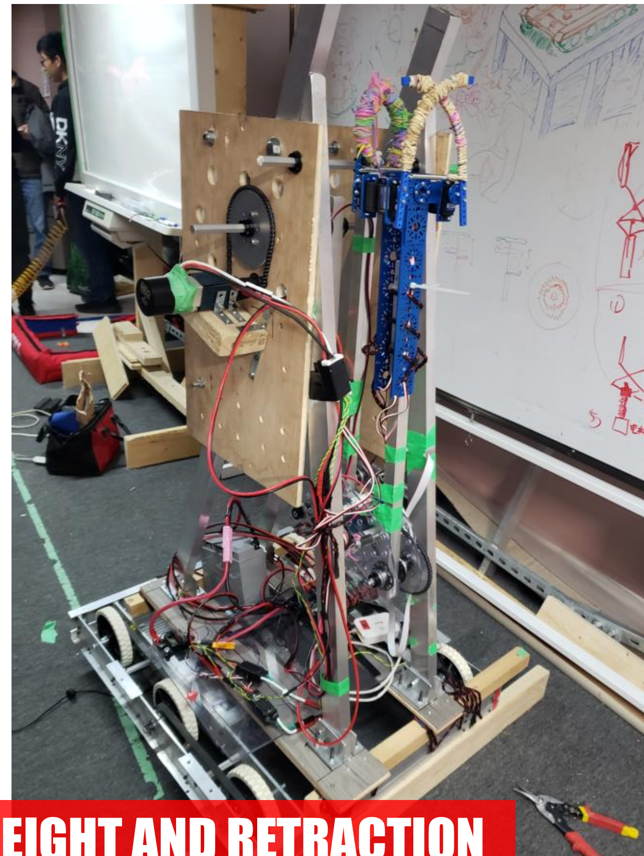
CAD-ING ELBOW PLATES TO BE CNC-MACHINED

IMPLEMENTATION AND FUNCTIONAL TESTING



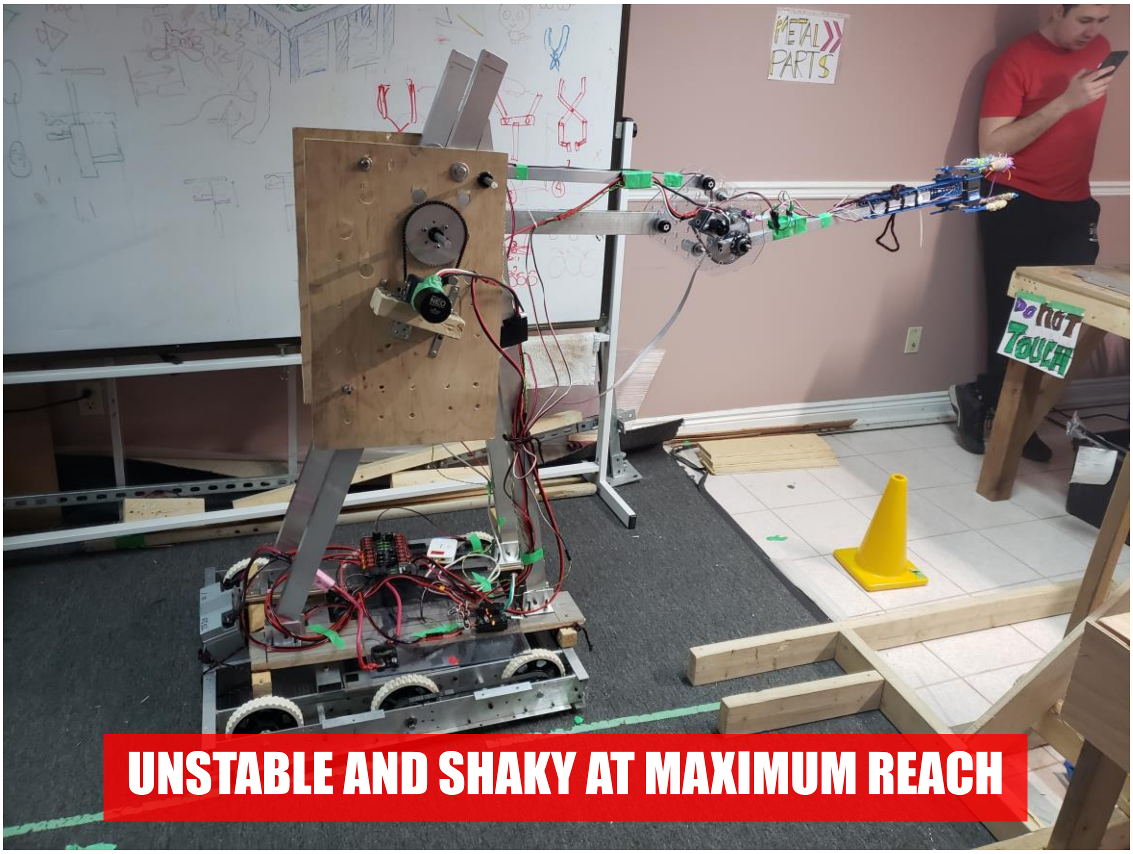
INTEGRATED METAL TORSO ON A TEST DRIVEBASE

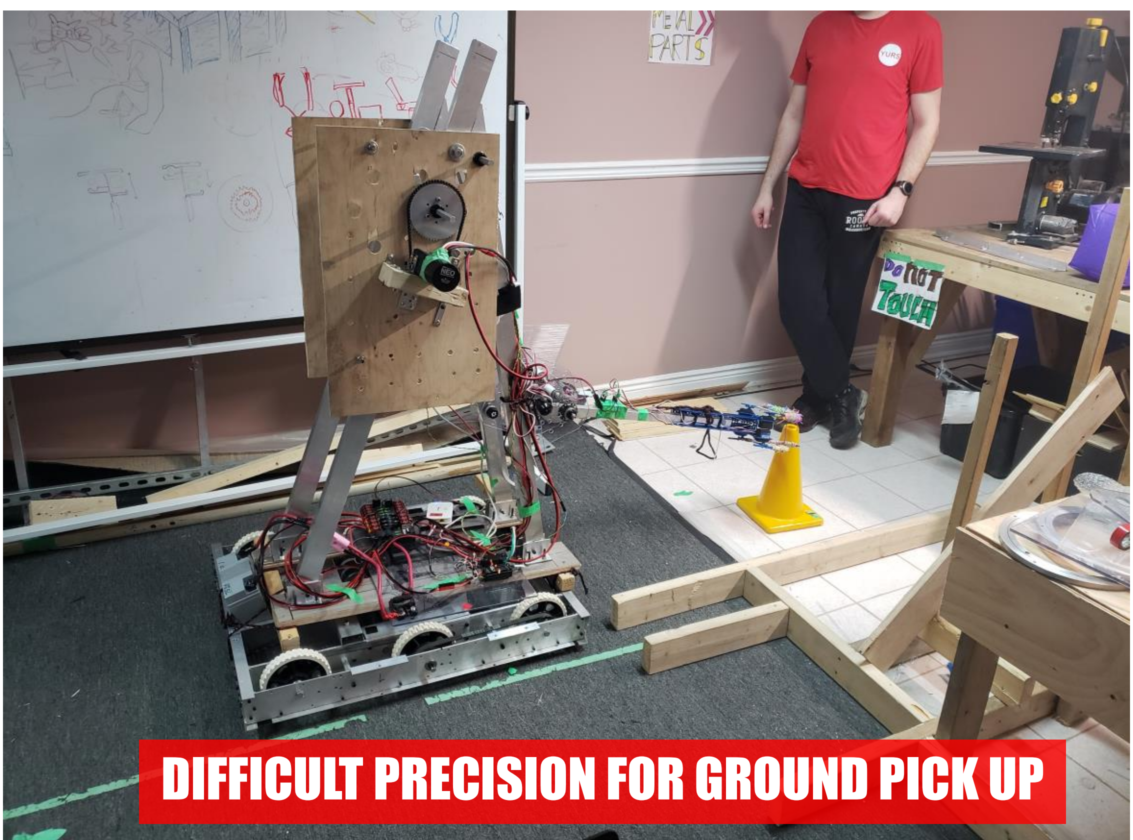


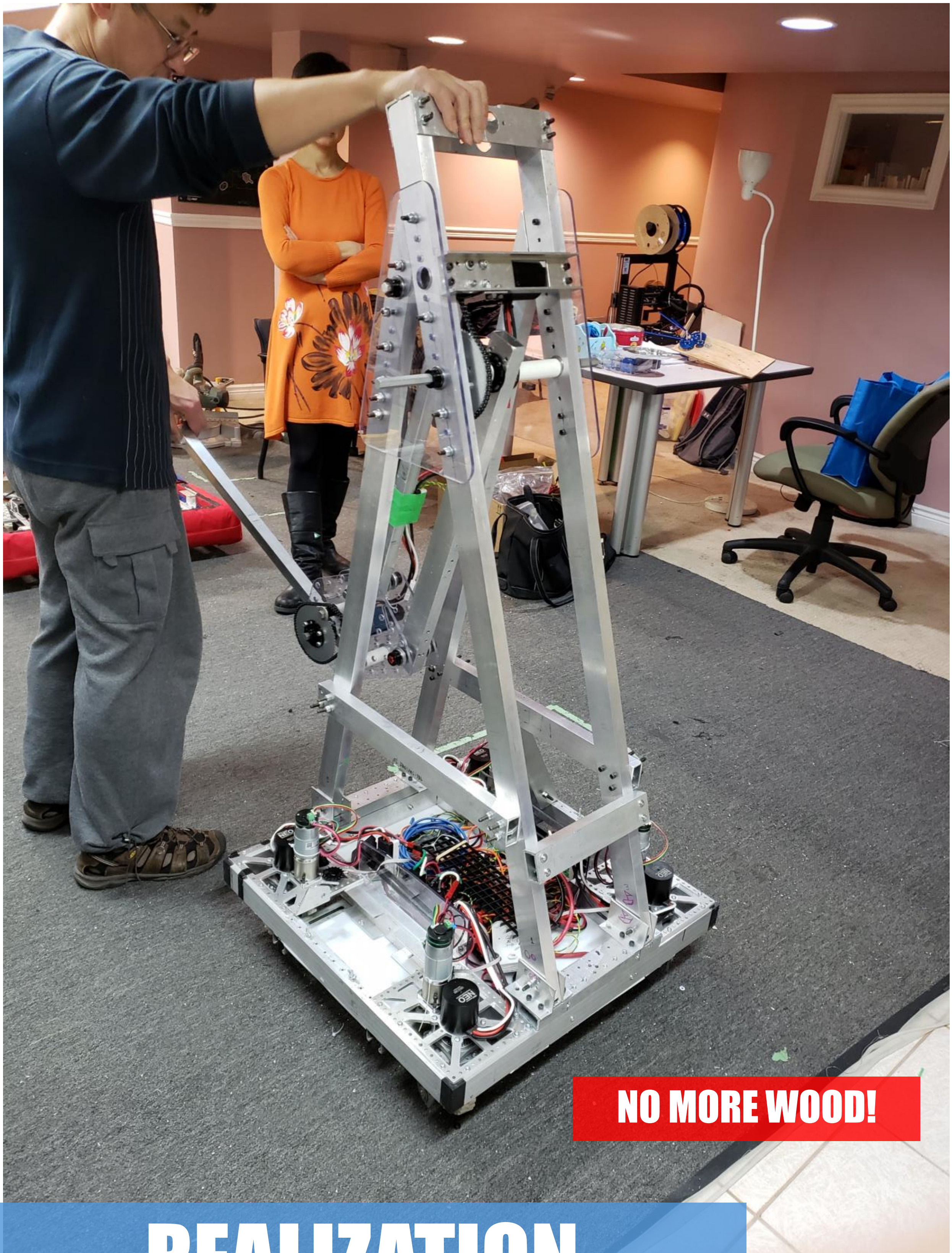


TESTING MAX HEIGHT AND RETRACTION

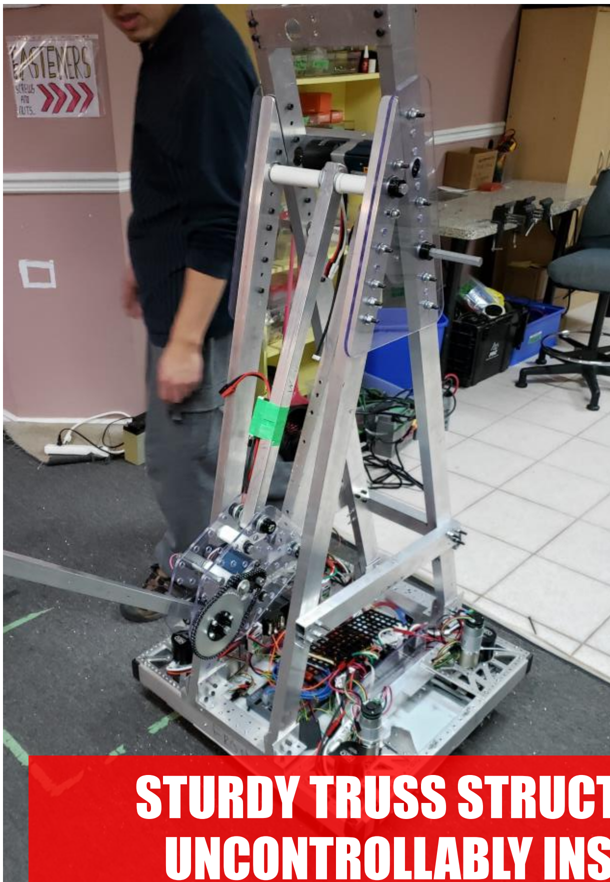
USER TESTING, TESTING GENERAL FUNCTIONALITY

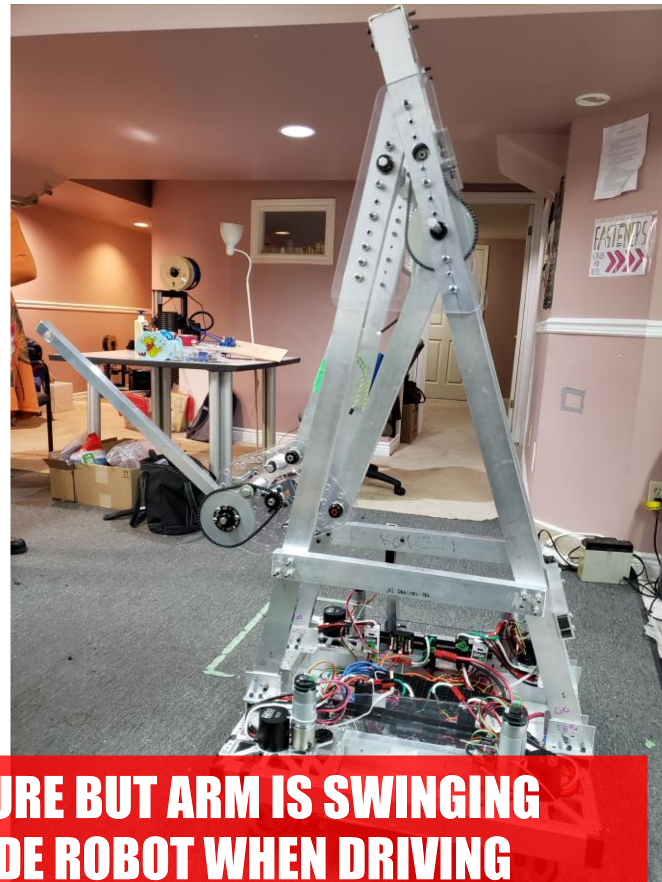

METAL PARTS
Do NOT TOUCH
UNSTABLE AND SHAKY AT MAXIMUM REACH


METAL PARTS
Do NOT TOUCH
DIFFICULT PRECISION FOR GROUND PICK UP



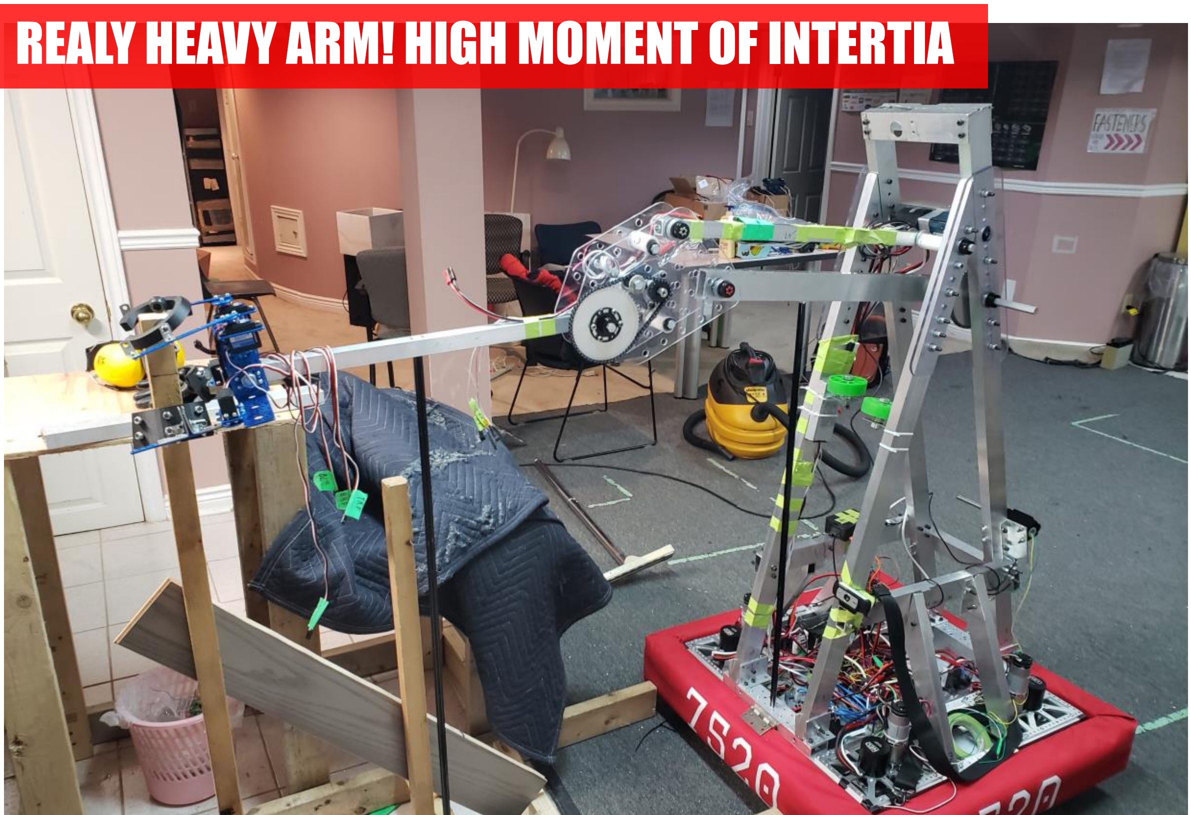
REALIZATION





STURDY TRUSS STRUCTURE BUT ARM IS SWINGING UNCONTROLLABLY INSIDE ROBOT WHEN DRIVING

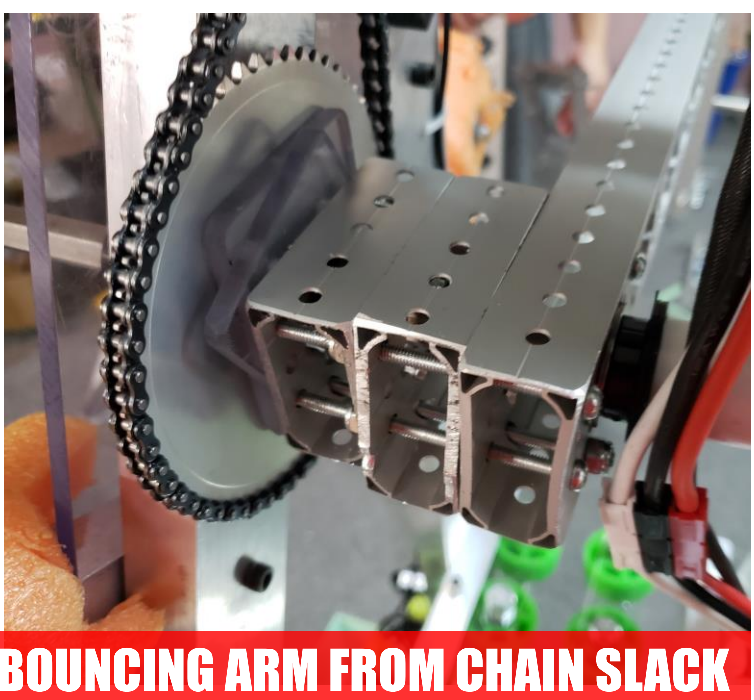
REALY HEAVY ARM! HIGH MOMENT OF INTERTIA



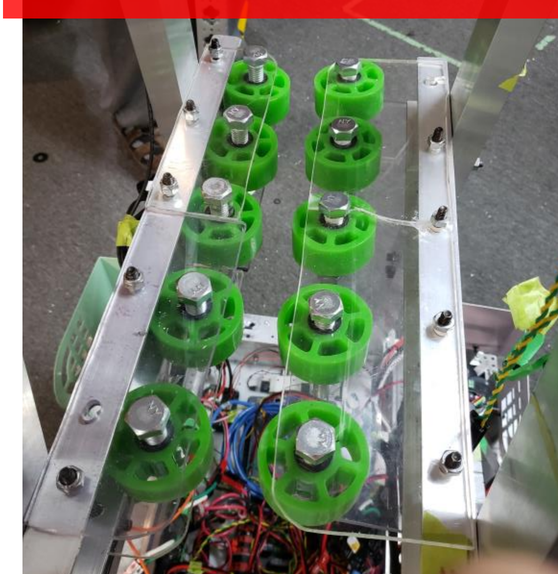
NON-FUNCTIONAL TESTING

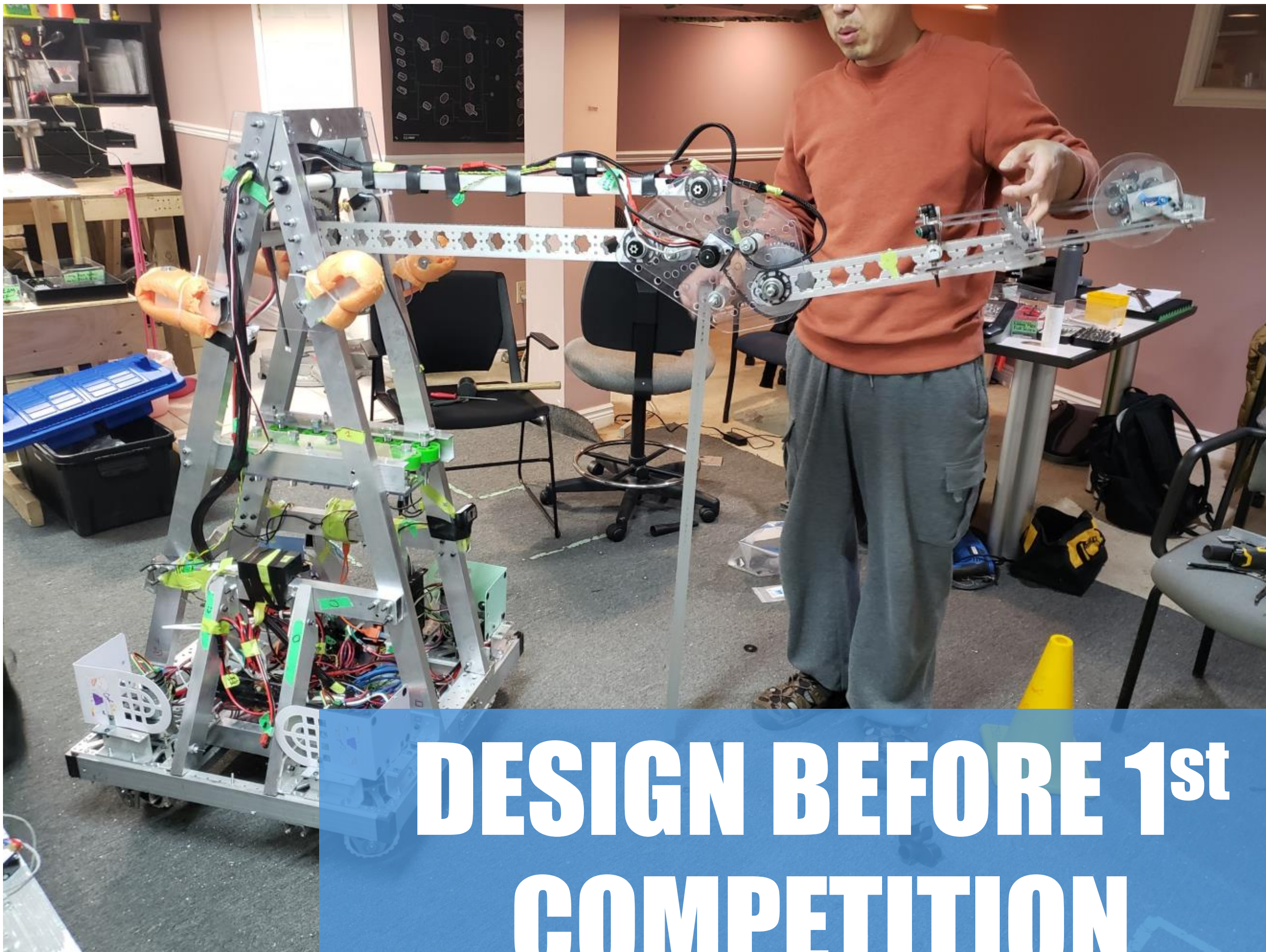

SHAFT DEFORMATION DUE TO HIGH TORQUE


HIGH CENTER OF MASS, VERY TIPPY


BOUNCING ARM FROM CHAIN SLACK

RUBBER WHEELS TO PREVENT SHAKING ARM





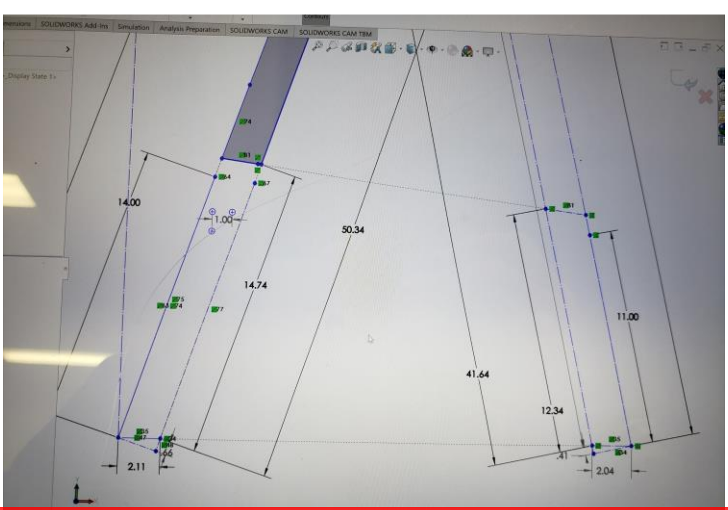
DESIGN BEFORE 1st COMPETITION

SWITCHING TO SWISS-CHEESED ALUMINUM FOR LIGHTER ARM

MOUNTING 4th ITERATION HAND

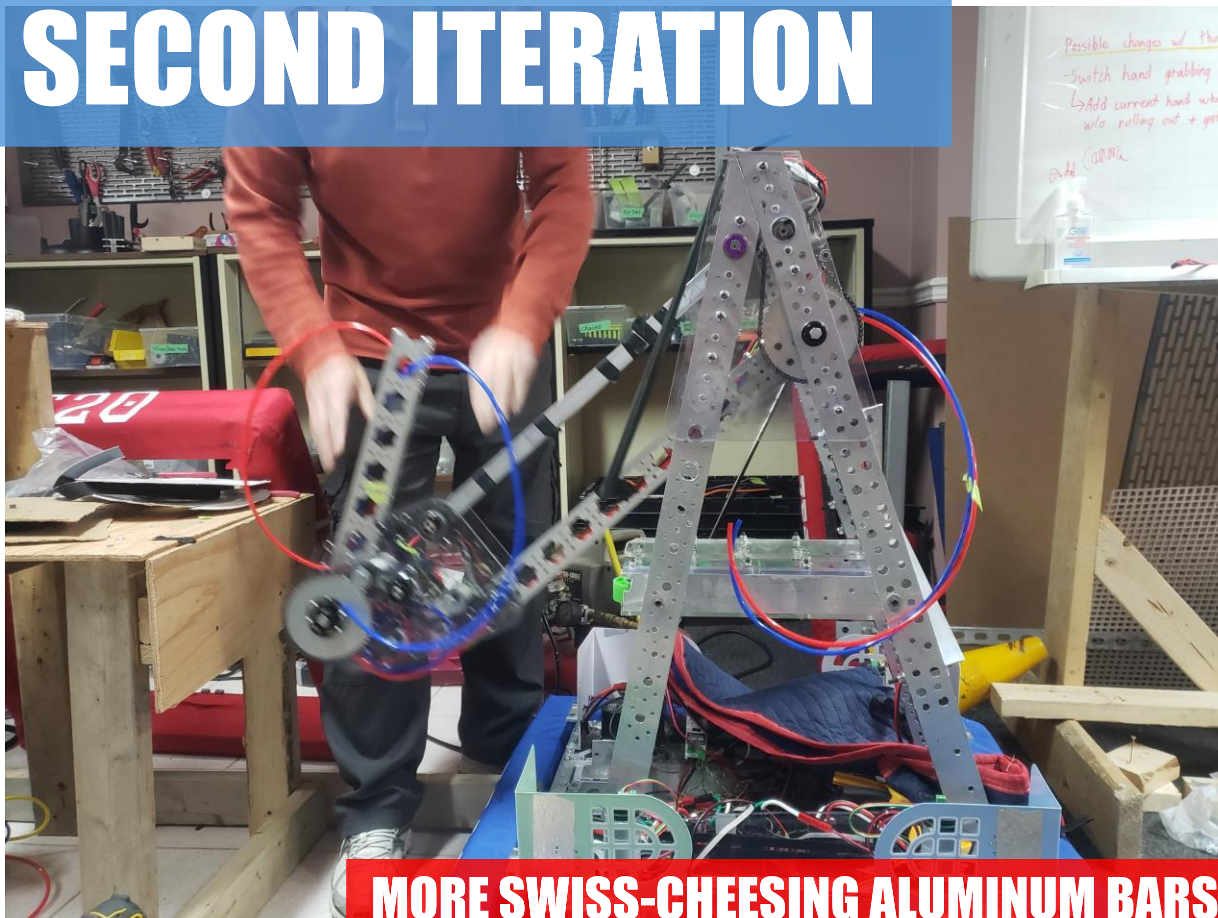
U WATERLOO COMPETITION

TOO TIPPY! TOO SHAKY! CANNOT DRIVE AT HIGH SPEEDS & THEREFORE INEFFICIENT



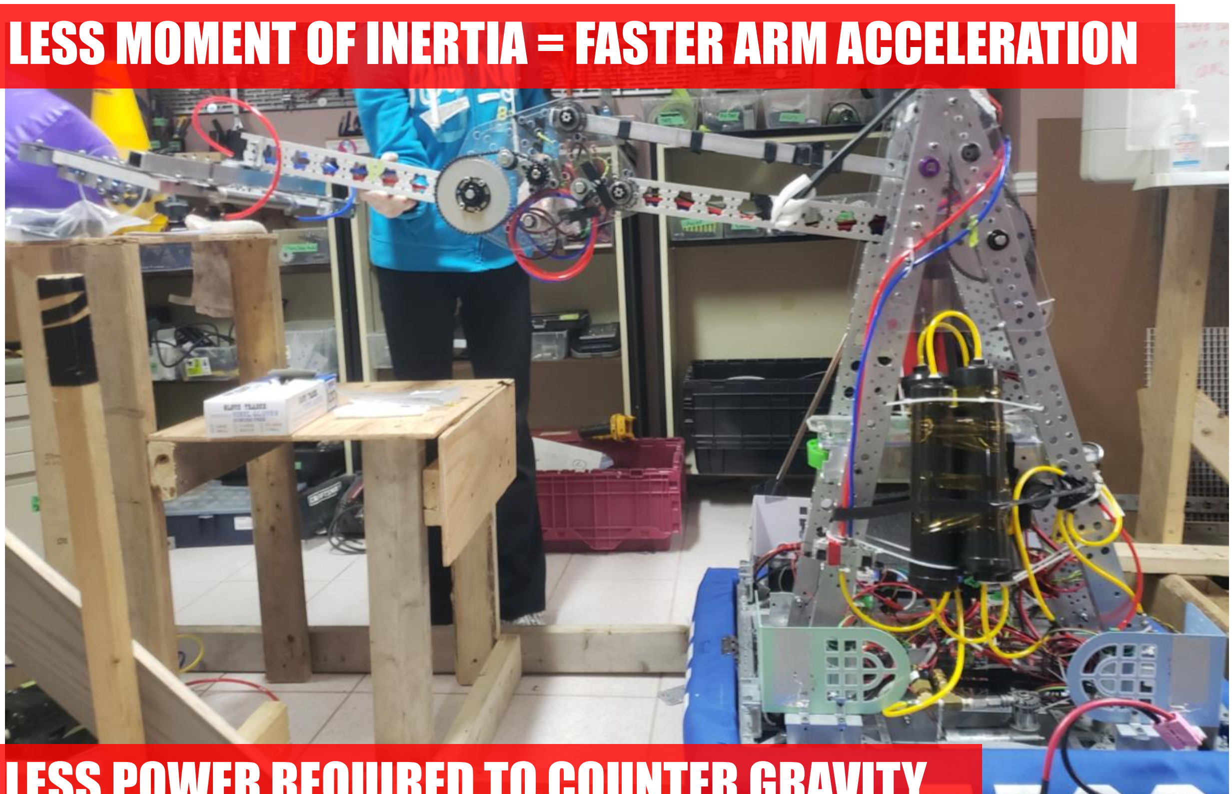
REDUCING ROBOT HEIGHT AND CENTER OF MASS

SECOND ITERATION

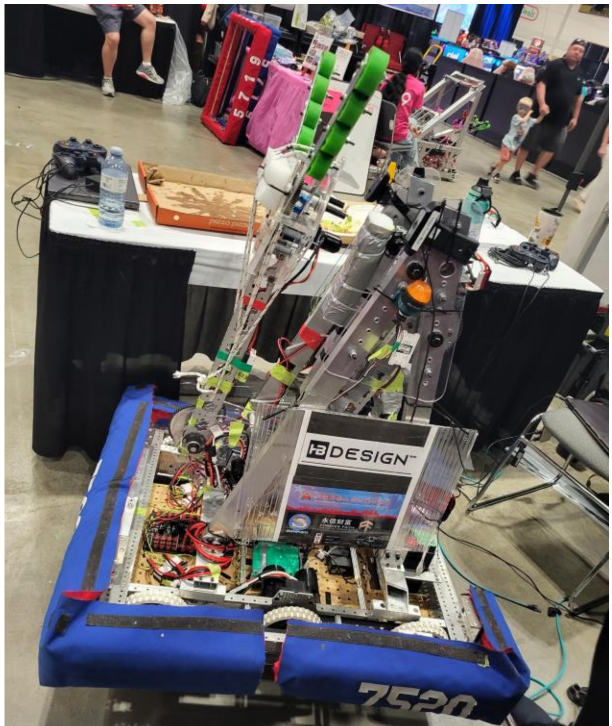


MORE SWISS-CHEESING ALUMINUM BARS

LESS MOMENT OF INERTIA = FASTER ARM ACCELERATION



LESS POWER REQUIRED TO COUNTER GRAVITY



THIRD ITERATION

IMPLEMENTATION CHALLENGES

ARM CALIBRATION – ARM ORIENTATION MUST BE KNOWN AT ALL TIMES DUE TO MECHANICAL RESTRICTIONS, BUT CALIBRATION IS TOO SLOW FOR A TIME-RESTRICTED SETTING. HOW CAN THE ROBOT DETERMINE ITS STATE AND MOVE SUCH THAT IT DOESN'T BREAK ITSELF?

INVERSE KINEMATICS – HOW SHOULD THE CLOSED LOOP FEEDBACK SYSTEM (PID) ROTATE MOTORS SUCH THAT THE ARM REACHES A SET POSITION VIA A CERTAIN TIME, PATH, AND ORIENTATION?

AUTONOMOUS MOVEMENT – DIFFICULTY MANUALLY PICKING UP AND PLACING OBJECTS IN RAPIDLY CHANGING ENVIRONMENTS AND OBSTRUCTED VIEWS REQUIRES IMPLEMENTATION OF SENSORS AND CAMERAS, BUT HOW WILL THE ROBOT KNOW IT HAS PERFORMED THE TASK SUCCESSFULLY?

WORK IN PROGRESS!

STILL MORE TO ADD HERE.

PLEASE CHECKOUT
THE 2024 ROBOT PROJECT HERE:

https://youtu.be/2KvQ7BLHxaI